

Convert Measurement Units

Lesson 4-6

Name: _____ Date: _____ Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



units of length

1 foot = 12 inches, so the factor is 12

Conversion Factor

Write the definition:

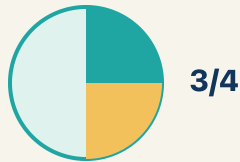


units of length

Inches, feet, pounds, ounces

Unit

Write the definition:



$\frac{3}{4}$

2 feet = 24 inches

Equivalent

Write the definition:



$3 : 2$

12 inches per 1 foot

Ratio

Write the definition:



units of length

*12 inches = 1 foot, 3 feet = 1 yard, 16 ounces = 1
pound*

Customary units

Write the definition:



units of length

*1 meter = 100 centimeters, 1 kilogram = 1,000
grams*

Metric units

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can convert measurement units using ratios and conversion factors.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Conversion Factor

Unit

Equivalent

Ratio

Customary units

Metric units

1 A ratio equal to 1 that relates two units, like 12 inches / 1 foot, is a _____.

2 A standard amount used to measure, like a foot or a gram, is a _____.

3 Two measurements that name the same amount, like 1 foot and 12 inches, are _____.

4 A comparison of two quantities is a _____.

5 Units like inches, feet, and pounds used mainly in the U.S. are _____.

6 Units like meters, grams, and liters based on tens are _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: How many inches are in 7 feet?

- A. 84 inches
- B. 72 inches
- C. 70 inches
- D. 96 inches

Step 1 $7 \times 12 = 84$ inches.

 **Answer:** A. 84 inches

 We do — together

Solve this one with your class using the same steps.

Problem: How many feet are in 3 yards?

- A. 9 feet
- B. 6 feet
- C. 12 feet
- D. 36 feet

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: How many ounces are in 4 pounds?

- A. 64 ounces
- B. 48 ounces
- C. 32 ounces
- D. 16 ounces

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Leg 3: The coin box weighs 5 pounds, but the shipping label needs the weight in ounces. How many ounces is that? (1 lb = 16 oz)

- A. 80 ounces
- B. 21 ounces
- C. 75 ounces
- D. 16 ounces

Show your work:

2. Final Leg: The finished cabinet ships in a crate holding 8 quarts of packing gel, but the supplier sells gel by the gallon. How many gallons is that? (4 qt = 1 gal)

- A. 2 gallons
- B. 32 gallons
- C. 4 gallons
- D. 12 gallons

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A recipe says you need 5 feet of ribbon, but the store only sells ribbon in inches. Another recipe says you need 2 yards of fabric, but your ruler measures in feet. Convert both measurements and explain the steps you used. Which conversion factor did you use each time?

Sentence starter: 5 feet = ____ inches because ____ . 2 yards = ____ feet because ____ . The conversion factors I used were ____ .

Show your work:

Reflect — Exit Ticket

Which is the correct conversion of 5 yards to inches?

- A. 180 inches
- B. 60 inches
- C. 15 inches
- D. 540 inches

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Conversion Factor
2. **Notes 2:** Unit
3. **Notes 3:** Equivalent
4. **Notes 4:** Ratio
5. **Notes 5:** Customary units
6. **Notes 6:** Metric units
7. **We Try (notes):** A. 9 feet — $1 \text{ yard} = 3 \text{ feet}$. $3 \times 3 = 9 \text{ feet}$.
8. **You Try (notes):** A. 64 ounces — $1 \text{ pound} = 16 \text{ ounces}$. $4 \times 16 = 64 \text{ ounces}$.
9. **Try It 1:** A. 80 ounces — *Ounces are smaller than pounds, so multiply: $5 \times 16 = 80 \text{ ounces}$.*
10. **Try It 2:** A. 2 gallons — *Gallons are bigger than quarts, so divide: $8 \div 4 = 2 \text{ gallons}$.*
11. **Exit Ticket:** A. 180 inches — $5 \text{ yards} \times 3 = 15 \text{ feet}$. $15 \text{ feet} \times 12 = 180 \text{ inches}$. Or: $5 \times 36 = 180 \text{ inches}$.